

Project 2: Image Stitching

張晴昀 (B10902064)

黃允謙 (B10902067)

May 1, 2025

1 Description

This project aims to stitch a set of images into a seamless panorama. The process involves several key image processing steps, including feature detection, description, and matching. Once corresponding features are matched across images, further operations such as warping, blending, alignment and cropping are performed to generate the final panoramic output. Image stitching enhances both the effective resolution and the field of view, making it a powerful technique for producing wide-angle visual representations.

2 Processing Flow

1. Feature Detection
 - Moravec Corner Detector
 - Harris Corner Detector (Bonus)
2. Feature Description
3. Feature Matching
4. Image matching
5. Blending
6. End-to-end Alignment (Bonus)
7. Rectangling (Bonus)

3 Feature Detection

3.1 Moravec Corner Detector

We first implement the simple Moravec corner detector. For each pixel, the algorithm calculates the intensity difference between its patch and a shifted version. Then, it uses the minimum difference among four shifted directions as the pixel's energy. A corner exhibits drastic intensity changes in all directions, and thus results in higher energy.

$$E(\mathbf{u}) = \sum_X w(\mathbf{x}) [I(\mathbf{x} + \mathbf{u}) - I(\mathbf{x})]^2$$

3.2 Harris Corner Detector

However, Moravec has several problems. First, it uses a binary window function for the patch, which makes it more susceptible to noise. Second, it only considers shifts in eight directions, potentially missing corners at other angles.

The Harris corner detector introduces an alternative method for computing the energy to address these issues. It uses a Gaussian function to produce a smoother window for the patch. Then, it applies Taylor expansion to approximate the intensity difference for a given shift direction, which can be expressed as:

$$E(\mathbf{u}) \approx \mathbf{u}^\top \mathbf{M} \mathbf{u}, \quad \mathbf{M} = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_y^2 & I_y^2 \end{bmatrix}$$

Since E is a quadratic function of the shift vector $\mathbf{u}$, we can analyze its shape by examining the properties of $\mathbf{M}$. As a result, Harris considered the eigenvalues of $\mathbf{M}$. When both eigenvalues are large, it indicates that E will be large in all directions $\mathbf{u}$, suggesting the presence of a corner. Harris proposed the following energy function:

$$E = \det \mathbf{M} - k(\text{tr} \mathbf{M})^2 = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$